

Savitzky-Golay optimizer diagnostics

Objective: mean squared translational/angular acceleration residual; angular residual uses the reference inertia/mass metric

Tolerances: {'ftol': None, 'xtol': None, 'gtol': 1e-06, 'note': 'ftol and xtol are disabled by default in the SG estimator; gtol or max_nfev terminates the solve unless explicitly overridden'}

Selected strict-ZOH solve

status=1 success=True nfev=4 njev=4

message='gtol' termination condition is satisfied.

reported optimality=5.775235756289021e-08

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initial={'cost': 1.5038903291662329, 'gradient_inf_norm': 0.04994451380746945,
'gradient_l2_norm': 0.06377722559589623, 'jacobian': {'condition_number_nonzero_subspace':
410.1917319583718, 'numerical_rank': 14, 'numerical_tolerance': 8.603465957419483e-11,
'singular_values': [119.51439259694365, 118.4739387034025, 102.4377954912419,
48.628227094280106, 44.72146406876796, 25.859831236989923, 12.98041888718595,
11.713440805485956, 11.328800522519675, 11.04511923545345, 9.85890907234415,
3.7972640495018757, 3.792575732655229, 0.291362266192807]}, 'residual_l2_norm':
1.7342954357122853}
final={'cost': 1.5038900847290744, 'gradient_inf_norm': 5.775235756289021e-08,
'gradient_l2_norm': 7.962116433089584e-08, 'jacobian': {'condition_number_nonzero_subspace':
410.4150611665243, 'numerical_rank': 14, 'numerical_tolerance': 8.603456067402414e-11,
'singular_values': [119.5142552105242, 118.47416683167468, 102.43782409332832,
48.628270457935415, 44.72148856823556, 25.85985499250417, 12.980399379817143,
11.713402233128225, 11.328807310040053, 11.045052961094457, 9.858860668236845,
3.7972768574031055, 3.792566334371872, 0.2912033853504995]}, 'residual_l2_norm':
1.7342952947690737}
coordinate_step_l2_norm=0.0002752809049669324
cost_reduction=2.44437158514188e-07
gradient_tolerance_satisfied=True
finite_difference_max_relative_error=8.195483990518276e-12
finite_difference_directional_checks=[{'analytic_l2_norm': 105.98239215125679,
'difference_l2_norm': 8.685769981545583e-10, 'direction': 'negative_gradient',
'finite_difference_l2_norm': 105.98239215151345, 'relative_error': 8.195483990518276e-12,
'scheme': 'central', 'step': 6.0554544523933395e-06}]
```

All smoothstep and strict-ZOH stage details are in optimizer_diagnostics.json.